Northwestern Kellogg

Northwestern ENGINEERING

MBAi 448 I Winter 2026

AI Today & Beyond

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Attendance

Enter the “Magic Word” in Canvas

Do Not Share with Absent Students

Image via ChatGPT

Our Judges

Armando Pauker
Co-Founder & Managing Director
Tensility Venture Partners

Ellie Bahrmasel
Founder
Stealth Mode

Erik Brown
Senior Director, Private Equity
Northwestern Kellogg

Oleg Evdokimov
Group Manager, Product Design
Relativity

Background Image via ChatGPT

The AI of Today

Image via ChatGPT

Everything Covered This Quarter

Image via ChatGPT

The AI of Tomorrow

Frontier Challenges and Potential Future Capabilities

Image via ChatGPT

What if AI could “perceive” and “understand” multiple modalities at once? Example Use Cases?


Image via ChatGPT

What if AI could “understand” and generate multiple modalities at once? Example Use Cases?

Your tools should help you work *better.*

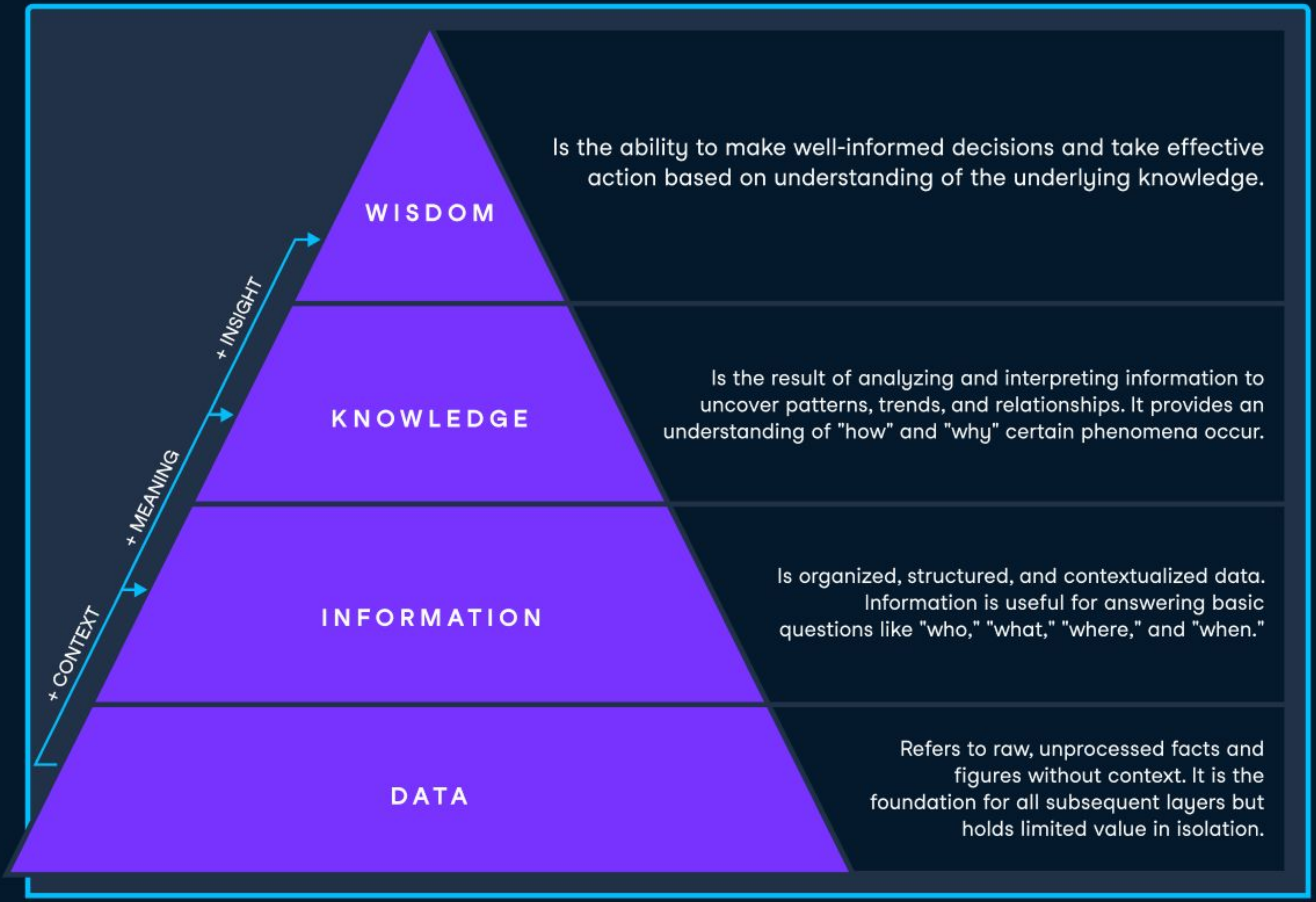
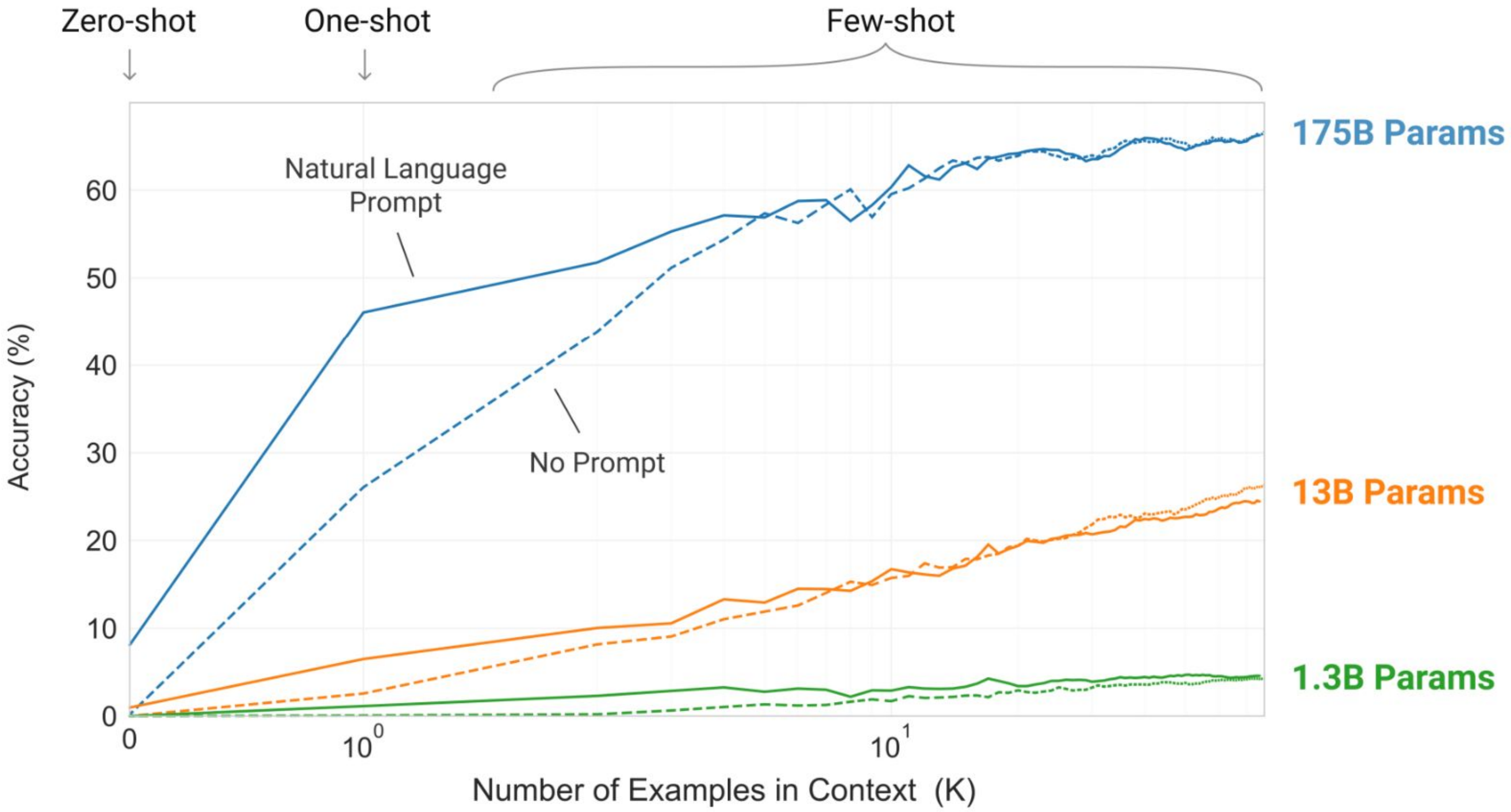


Image via Datacamp

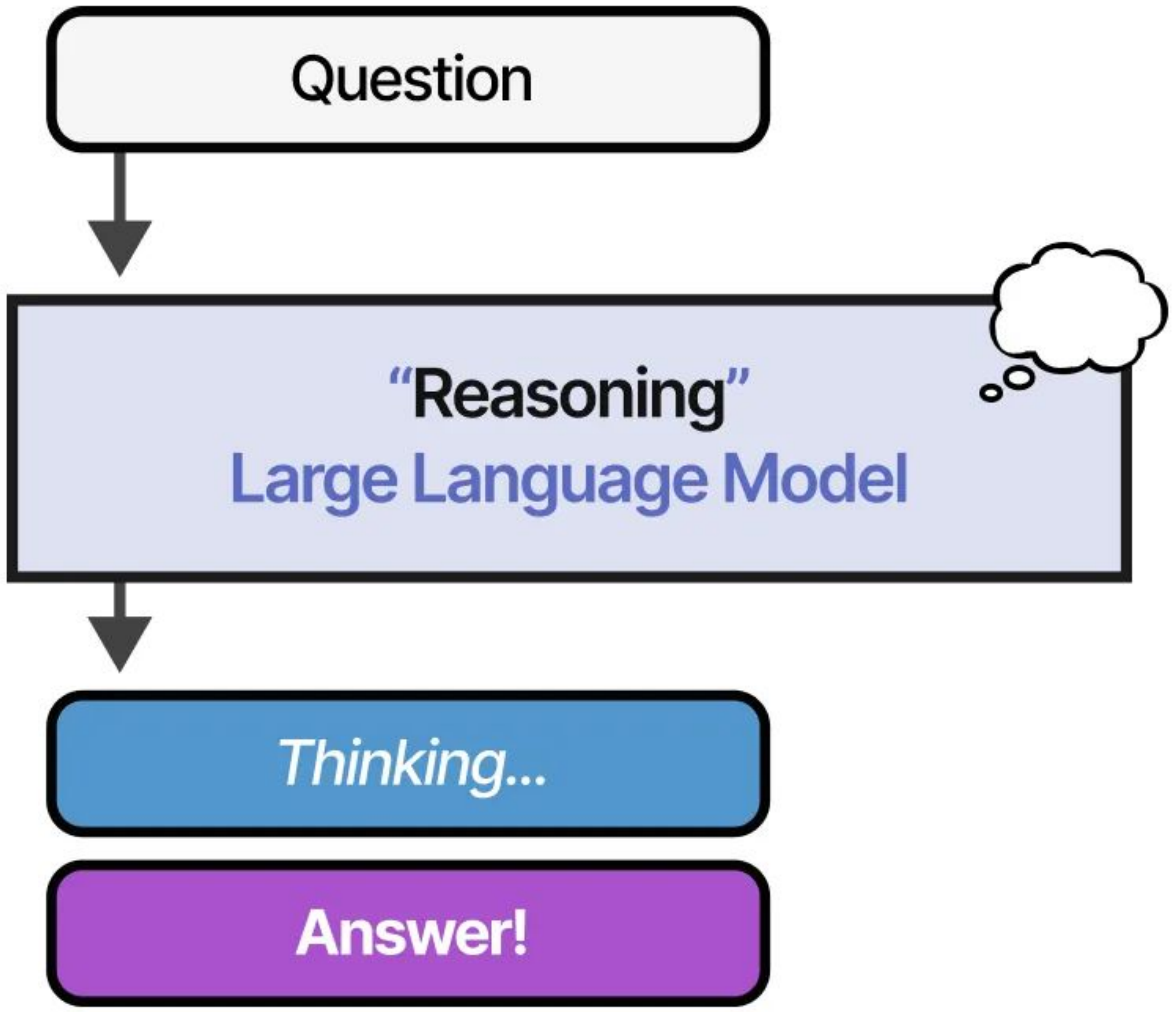
Scaling Deep Learning + Symbolic AI



Combining neural networks with logic-based reasoning to build AI systems that can handle both pattern recognition and explicit rules.

Source: Brown et al

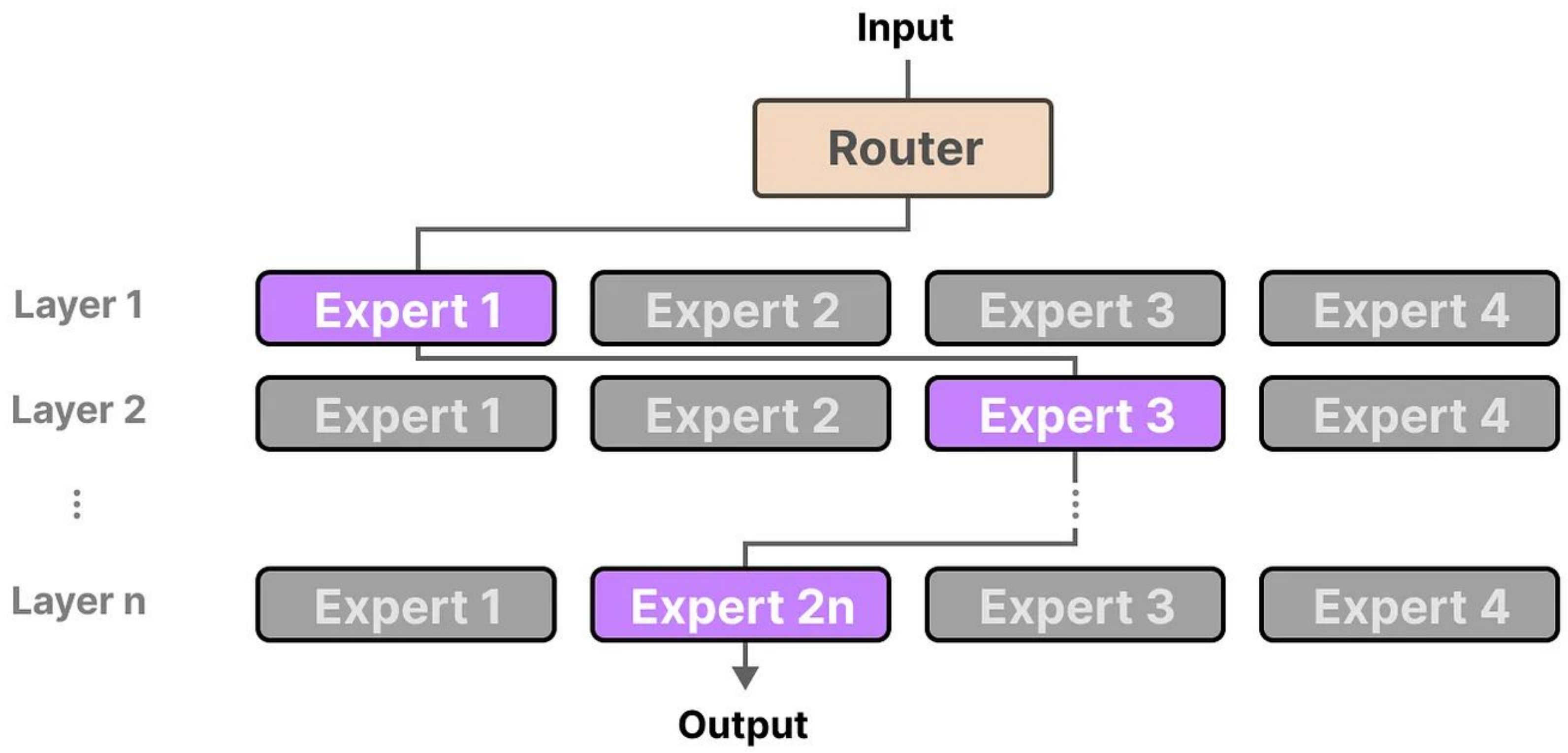
Chain of Thought & Reasoning



Prompting techniques that guide AI to show its reasoning process step-by-step, improving problem-solving accuracy and transparency.

Image via Maarten Grootendorst

Mixture of Experts



Mixture of Experts boosts AI performance by activating specialized models for specific tasks, improving efficiency and scalability.

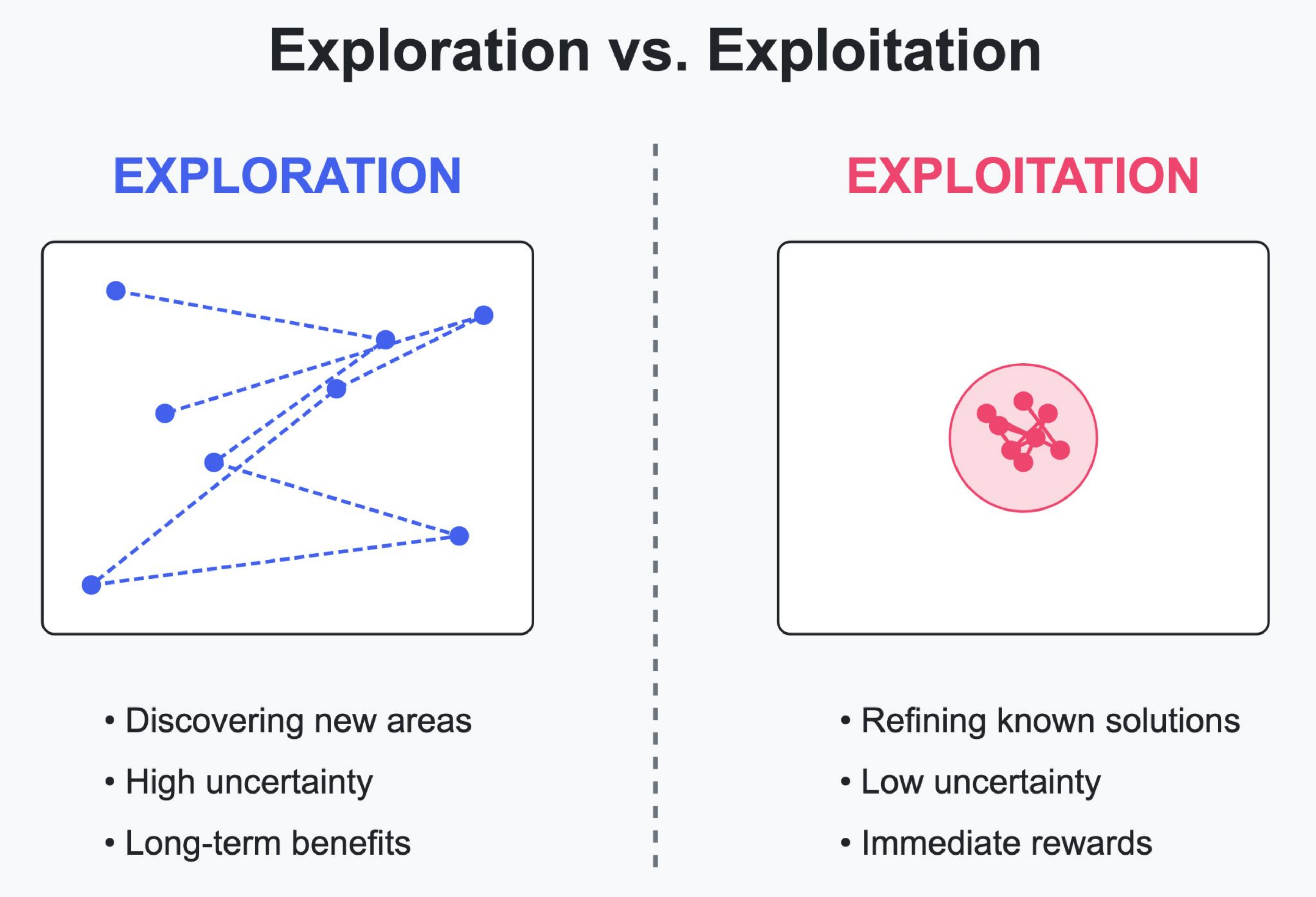
Image via Maarten Grootendorst

Knowledge Distillation & Small Models

Transferring knowledge from large AI models to smaller ones, creating efficient, faster systems that maintain quality while reducing computing needs.

Image via Gemini

Search, Sampling, and Entropy

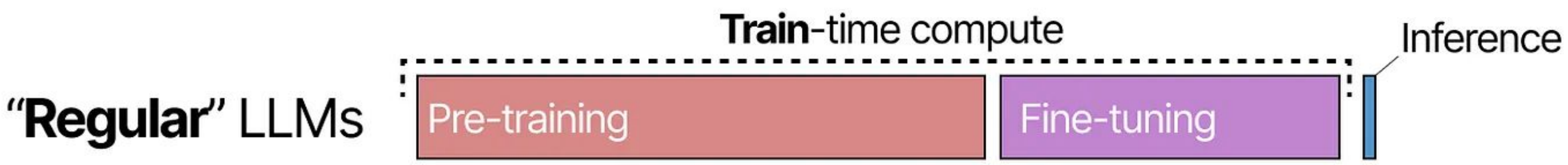


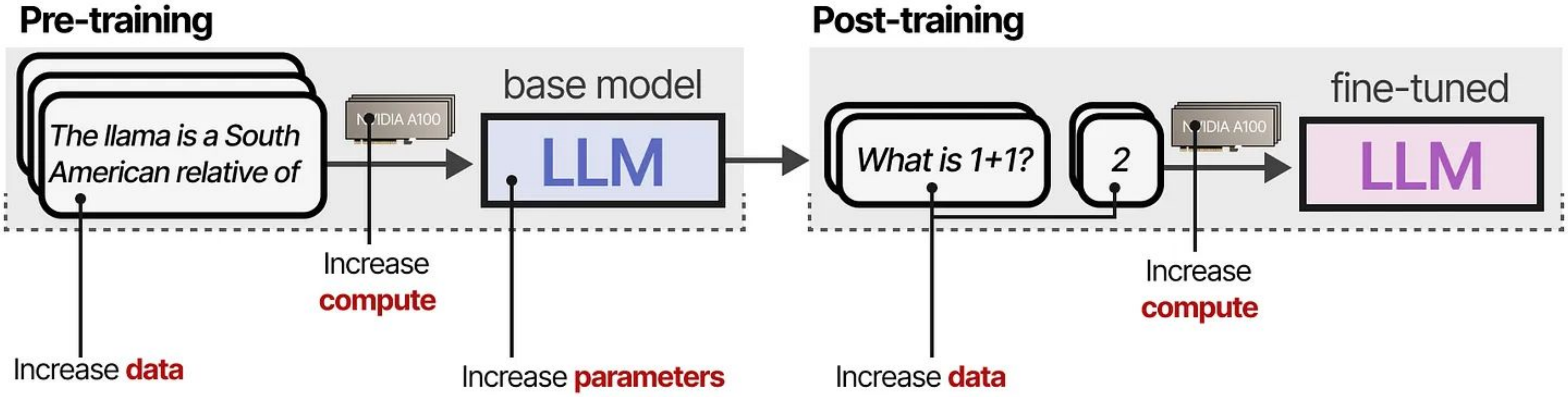
Algorithms that efficiently explore solution spaces, critical for helping AI systems gain general intelligence capabilities (e.g., Monte Carlo Tree Search and Beam Search).

Image via Claude

What’s the difference between learning and search?

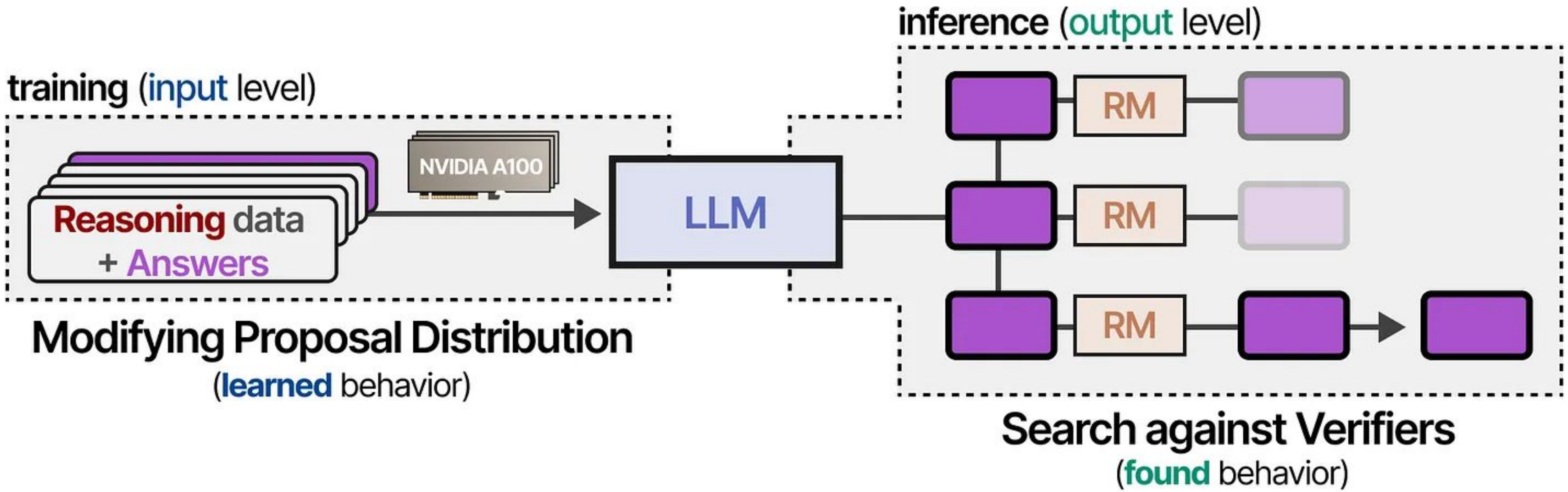
Training-time Compute





Images via Maarten Grootendorst

Test-time Compute



Images via Maarten Grootendorst

Curiosity, Inventive Creativity, & Novelty

Image via Gemini

Driving AI to explore new ideas, create unique solutions, and push beyond existing knowledge and capabilities.

Causality, Planning, & Agency

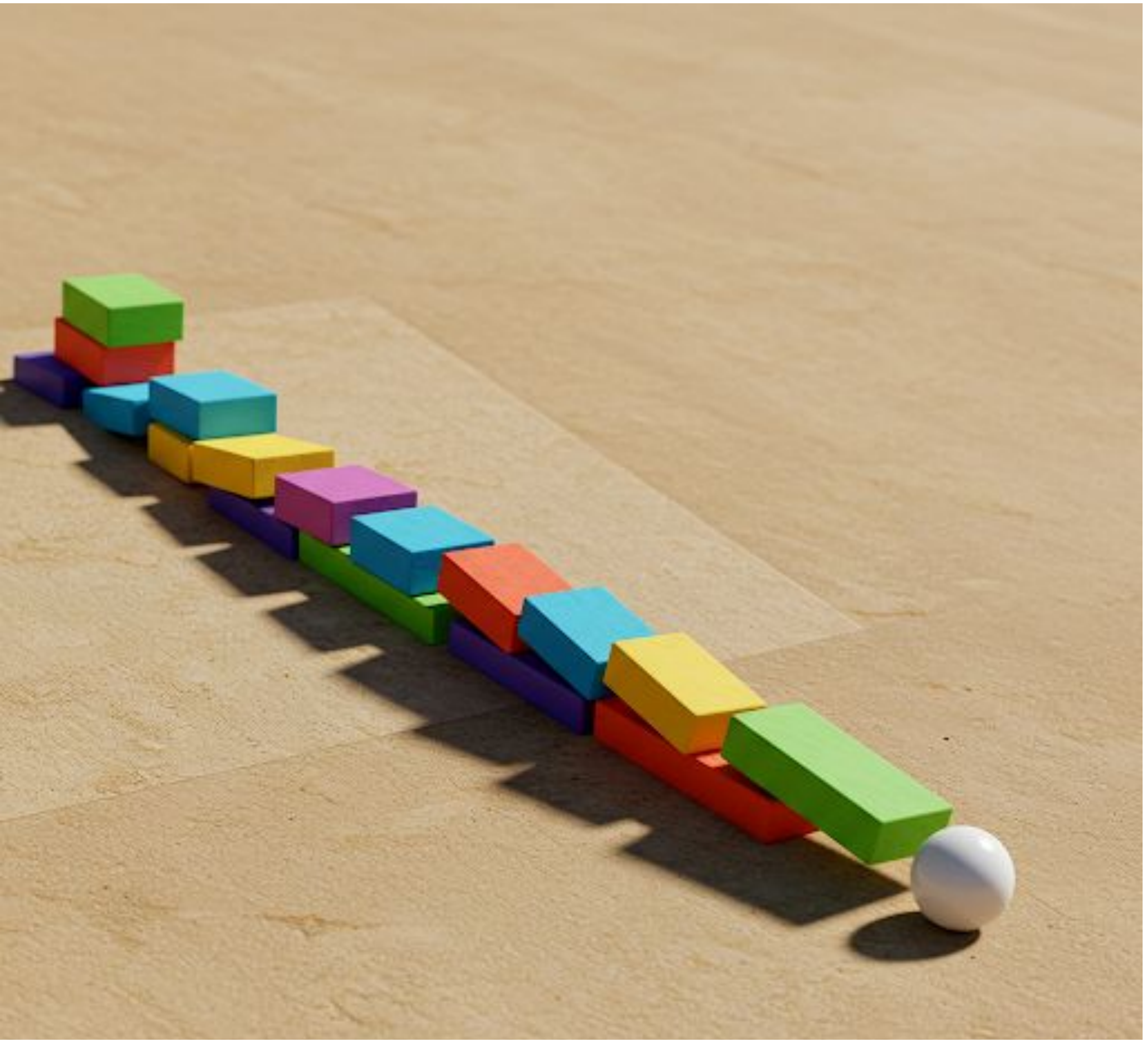
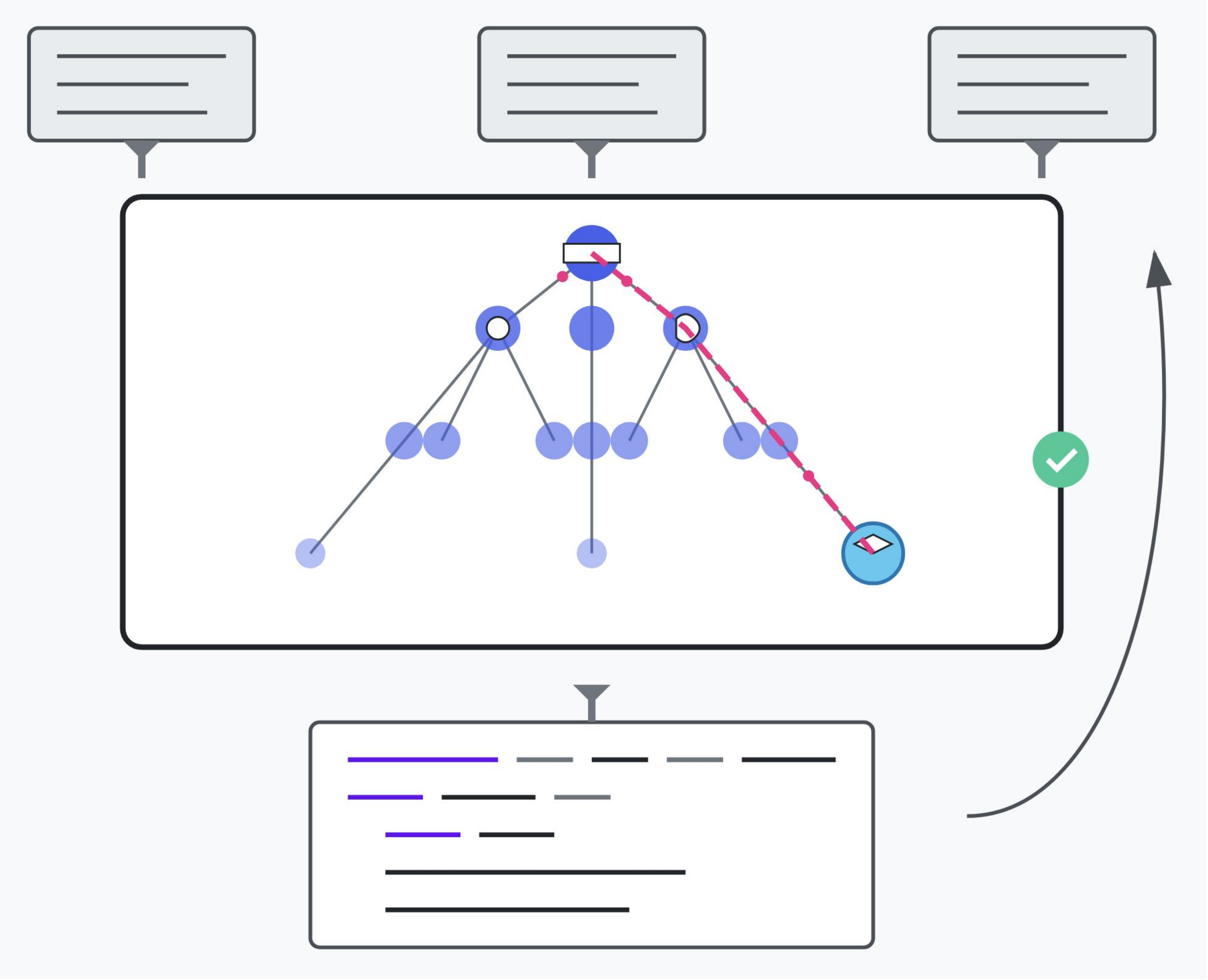


Image via Gemini

Helping AI understand cause-effect relationships, make strategic decisions, and act with goal-directed behavior.

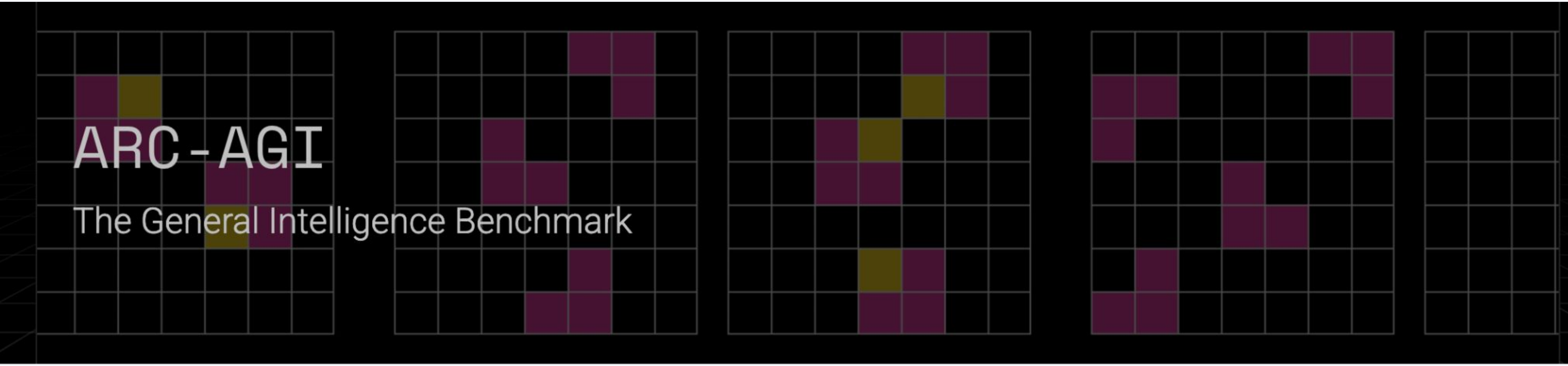
Abstractions & Synthesis



Enabling AI to simplify complex information and combine ideas, mimicking how humans understand and solve problems.

Image via Claude

AGI Definition per ARC-AGI



AGI is a system that can efficiently acquire new skills outside of its training data.

Easy for Humans, Hard for AI

Source:: ARC Prize

ARC PRIZE

ARC PRIZE REMAINS UNDEFEATED.
NEW IDEAS STILL NEEDED

OpenAI o3 unlocks breakthrough high score on ARC-AGI-Pub. Learn more.

ARC PRIZE

ARC Prize is a $1,000,000+ public competition to beat and open source a solution to the ARC-AGI benchmark.

Hosted by Mike Knoop (Co-founder, Zapier) and François Chollet (Creator of ARC-AGI, Keras).

ARC Prize 2025 coming soon.

› See 2024 winners

ARC-AGI

Most AI benchmarks measure skill. But skill is not intelligence. General intelligence is the ability to efficiently acquire new skills. Chollet's unbeaten 2019 Abstraction and Reasoning Corpus for Artificial General Intelligence (ARC-AGI) is the only formal benchmark of AGI progress.

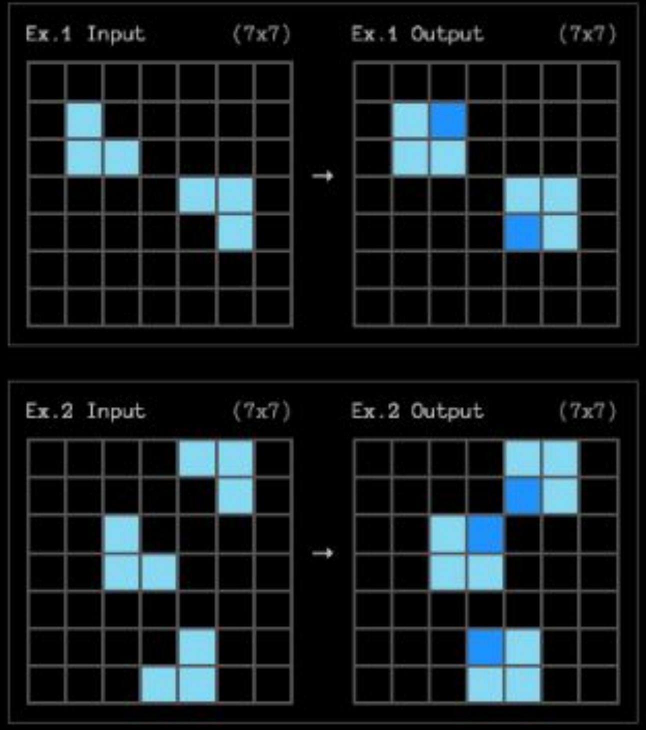
It's easy for humans, but hard for AI.

PLAY

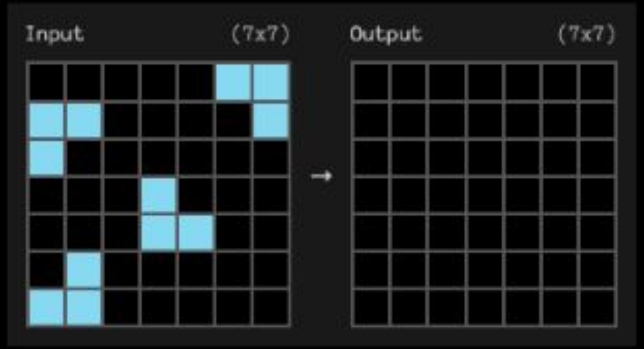
Try ARC-AGI. Given the examples, identify the pattern and solve the test puzzle.

Puzzle ID: 3aa6fb7a   Previous 1 of 6 Next

EXAMPLES



TEST



1. Configure your output grid:

Copy from input Reset

2. Click to select a color:

3. See if your output is correct:

Submit solution

Demo

Feel free to follow along!

Artificial General Intelligence (AGI)

Goals:

- Explore the ARC-AGI Benchmark and ARC-AGI Prize to learn more about the current progress towards AGI

Links:

- ARC Prize

Reflect + Discuss:

- Share your thoughts

Just A Rather Very Intelligent System

Source: Marvel Cinematic Universe Wiki

Researching Emerging & State-of-the-Art AI

Artificial Intelligence

Authors and titles for recent submissions

- Wed, 21 Aug 2024
- Tue, 20 Aug 2024
- Mon, 19 Aug 2024
- Fri, 16 Aug 2024
- Thu, 15 Aug 2024

See today's new changes

Total of 611 entries : 1-25 26-50 51-75 76-100 ... 601-611
Showing up to 25 entries per page: fewer | more | all

Wed, 21 Aug 2024 (showing first 25 of 164 entries)

[1] arXiv:2408.11042 [pdf, html, other]

GraphFSA: A Finite State Automaton Framework for Algorithmic Learning on Graphs
Florian Grötschla, Joël Mathys, Christoffer Raun, Roger Wattenhofer
Comments: Published as a conference paper at ECAI 2024
Subjects: **Artificial Intelligence (cs.AI)**

[2] arXiv:2408.11039 [pdf, html, other]

Transfusion: Predict the Next Token and Diffuse Images with One Multi-Modal Model
Chunting Zhou, Lili Yu, Arun Babu, Kushal Tirumala, Michihiro Yasunaga, Leonid Shamis, Jacob Kahn, Xuezhe Ma, Luke Zettlemoyer, Omer Levy
Comments: 23 pages
Subjects: **Artificial Intelligence (cs.AI)**; Computer Vision and Pattern Recognition (cs.CV)

[3] arXiv:2408.10970 [pdf, html, other]

Hybrid Recurrent Models Support Emergent Descriptions for Hierarchical Planning and Control
Poppy Collis, Ryan Singh, Paul F Kinghorn, Christopher L Buckley
Comments: 4 pages, 3 figures
Subjects: **Artificial Intelligence (cs.AI)**; Systems and Control (eess.SY)

[4] arXiv:2408.10947 [pdf, html, other]

Dr.Academy: A Benchmark for Evaluating Questioning Capability in Education for Large Language Models
Yuyan Chen, Chenwei Wu, Songzhou Yan, Panjun Liu, Haoyu Zhou, Yanghua Xiao

The Thirty-eighth Annual Conference on Neural Information Processing Systems

NeurIPS 2024

Vancouver, Canada Dec 16 2024 https://neurips.cc/ pc2024@neurips.cc

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Please see our call for papers with detailed instructions before you submit.
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Please join the NeurIPS 2024 Checklist Assistant Study that will provide you with free verification of your checklist performed by an LLM: form link, see our blog for details.
Submission Start: Apr 22 2024 08:00PM UTC-0, Abstract Registration: May 15 2024 08:00PM UTC-0, Submission Deadline: May 22 2024 08:00PM UTC-0

Accept (oral) | Accept (spotlight) | Accept (poster) | Reject | Recent Activity

Learning to grok: Emergence of in-context learning and skill composition in modular arithmetic tasks
Tianyu He, Darshil Doshi, Aritra Das, Andrey Gromov
Published: 25 Sept 2024, Last Modified: 06 Nov 2024 NeurIPS 2024 oral Readers: Everyone 14 Replies
Show details

Trading Place for Space: Increasing Location Resolution Reduces Contextual Capacity in Hippocampal Codes
Spencer Rooke, Zhaoze Wang, Ronald W Di Tullio, Vijay Balasubramanian
Published: 25 Sept 2024, Last Modified: 10 Jan 2025 NeurIPS 2024 oral Readers: Everyone 11 Replies
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Visual Autoregressive Modeling: Scalable Image Generation via Next-Scale Prediction
Keyu Tian, Yi Jiang, Zehuan Yuan, BINGYUE PENG, Liwei Wang
Published: 25 Sept 2024, Last Modified: 16 Jan 2025 NeurIPS 2024 oral Readers: Everyone 30 Replies
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Cracking the Code of Juxtaposition: Can AI Models Understand the Humorous Contradictions
Zhe Hu, Tuo Liang, Jing Li, Yiren Lu, Yunlai Zhou, Yiran Qiao, Jing Ma, Yu Yin
Published: 25 Sept 2024, Last Modified: 06 Nov 2024 NeurIPS 2024 oral Readers: Everyone 26 Replies
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Human Expertise in Algorithmic Prediction
Rohan Alur, Manish Raghavan, Devavrat Shah
Published: 25 Sept 2024, Last Modified: 06 Nov 2024 NeurIPS 2024 oral Readers: Everyone 15 Replies
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Learning diffusion at lightspeed
Antonio Terpin, Nicolas Lanzetti, Martín Gadea, Florian Dorfler
Published: 25 Sept 2024, Last Modified: 06 Nov 2024 NeurIPS 2024 oral Readers: Everyone 13 Replies
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Weisfeiler and Leman Go Loopy: A New Hierarchy for Graph Representational Learning
Raffaele Paolino, Sohir Maskey, Pascal Welke, Gitta Kutyniok
Published: 25 Sept 2024, Last Modified: 06 Nov 2024 NeurIPS 2024 oral Readers: Everyone 12 Replies
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Unlocking the Capabilities of Thought: A Reasoning Boundary Framework to Quantify and Optimize Chain-of-Thought
Qiguang Chen, Libo Qin, Jiaqi WANG, Jingxuan Zhou, Wanxiang Che
Published: 25 Sept 2024, Last Modified: 25 Dec 2024 NeurIPS 2024 oral Readers: Everyone 23 Replies
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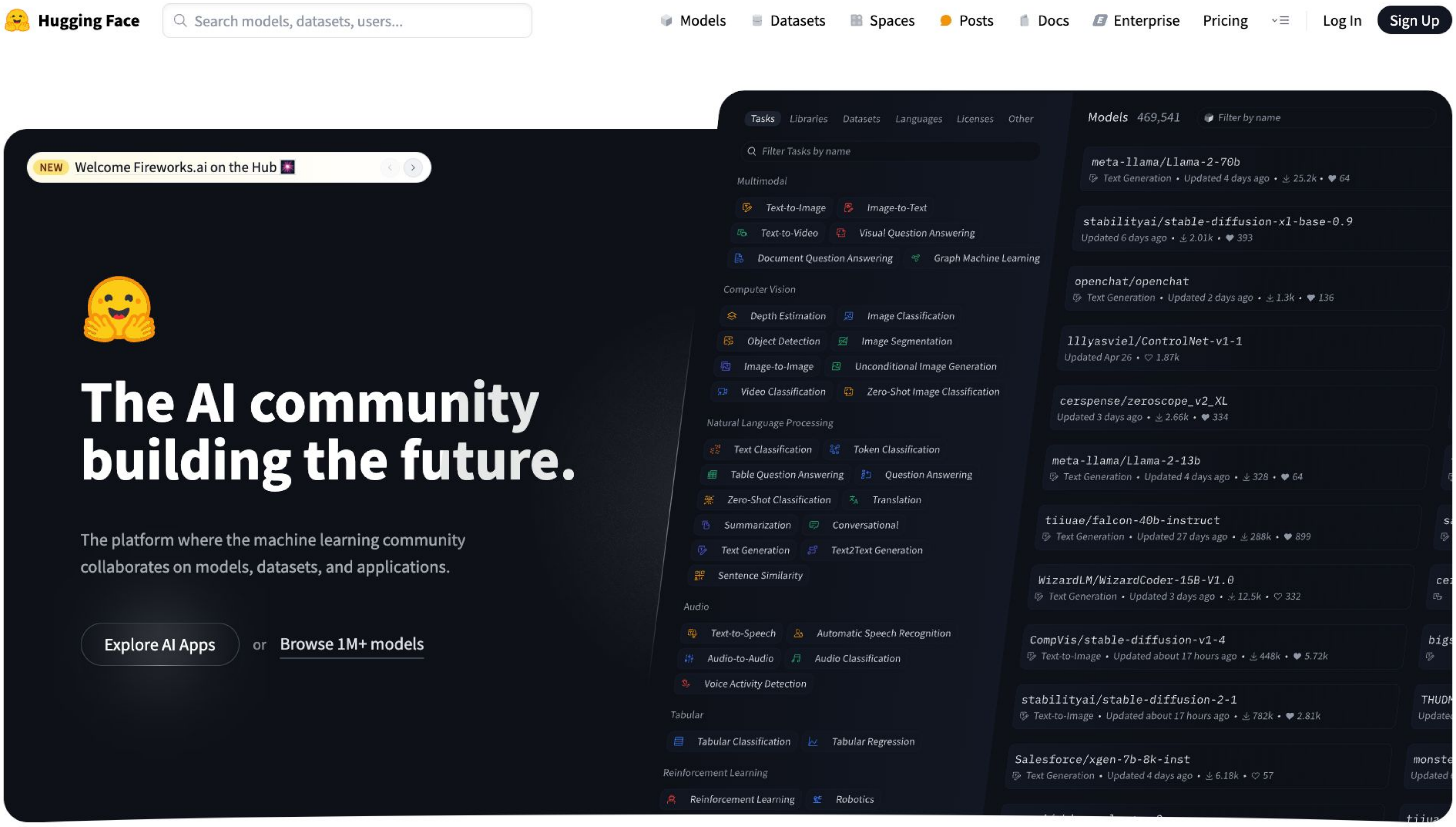

https://huggingface.co

kaggle

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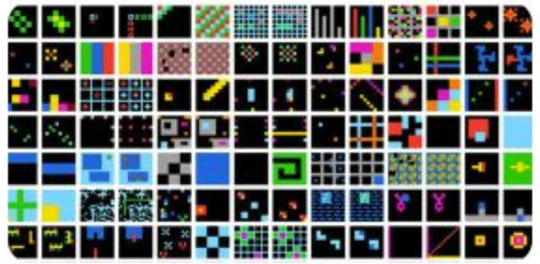
ABSTRACT AND REASONING CORPUS · FEATURED CODE COMPETITION · 3 MONTHS AGO

Late Submission

ARC Prize 2024

Create an AI capable of solving reasoning tasks it has never seen before



Overview Data Code Models Discussion Leaderboard Rules

Overview

In this competition, you'll develop AI systems to efficiently learn new skills and solve open-ended problems, rather than depend exclusively on AI systems trained with extensive datasets. The top submissions will show improvement toward human reasoning benchmarks.

Start
Jun 11, 2024

Close
Nov 10, 2024

Merger & Entry

Competition Host
Abstraction and Reasoning Corpus

Prizes & Awards
$1,100,000
Awards Points & Medals

Participation
16,349 Entrants
1,637 Participants
1,427 Teams
16,956 Submissions

https://www.kaggle.com

Keepin’ Up With AI

Image via ChatGPT

OpenAI

ANTHROP\C

DeepMind

Microsoft Research

Google AI

Meta

Hugging Face

Radar

Radar Trends

A monthly list of technology and business trends we find important or interesting.

Courses

Google Alerts

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Why of AI®

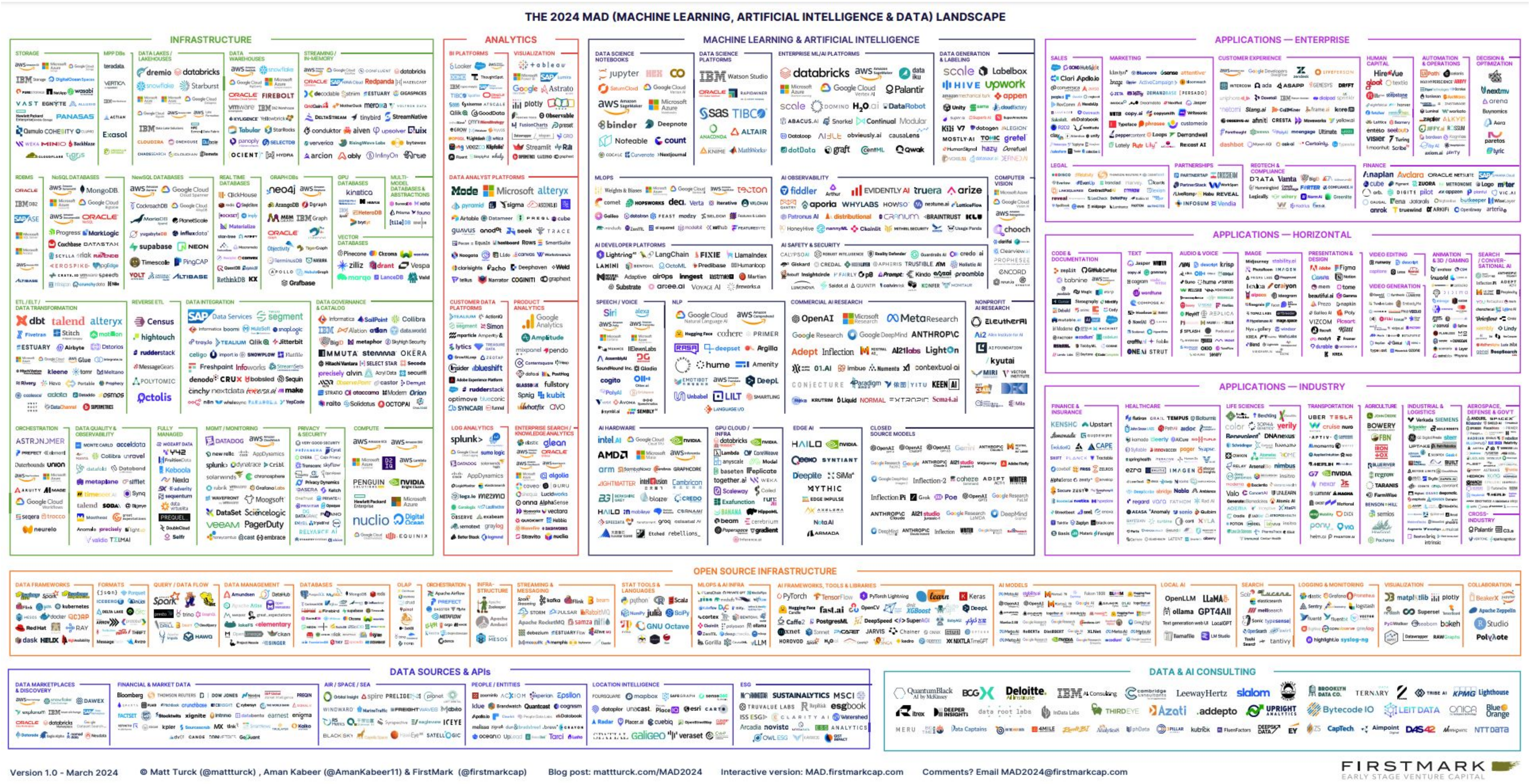

Image: Matt Turck

ANDREESSEN HOROWITZ

THE TOP 100 GEN AI CONSUMER APPS

4TH EDITION

2025

Product Hunt

GitHub

Google Cloud

Azure

aws

Demo

Feel free to follow along!

Staying Current

Goals:

- Explore sources and tools for staying current on the latest AI research and tools

Links:

- Hugging Face Trending Papers
- MAD Landscape (Full Resolution / Interactive)

Reflect + Discuss:

- Share your thoughts

How do you keep up with AI news or trends?

AI Predictions

Image via ChatGPT

10 predictions for the next 12 months

- **1.** A major retailer reports >5% of online sales from agentic checkout as AI agent advertising spend hits $5B.
- **2.** A major AI lab leans back into open-sourcing frontier models to win over the current US administration.
- **3.** Open-ended agents make a meaningful scientific discovery end-to-end (hypothesis, expt, iteration, paper).
- **4.** A deepfake/agent-driven cyber attack triggers the first NATO/UN emergency debate on AI security.
- **5.** A real-time generative video game becomes the year's most-watched title on Twitch.
- **6.** "AI neutrality" emerges as a foreign policy doctrine as some nations cannot or fail to develop sovereign AI.
- **7.** A movie or short film produced with significant use of AI wins major audience praise and sparks backlash.
- **8.** A Chinese lab overtakes the US lab dominated frontier on a major leaderboard (e.g. LMArena/Artificial Analysis).
- **9.** Datacenter NIMBYism takes the US by storm and sways certain midterm/gubernatorial elections in 2026.
- **10.** Trump issues an executive order to ban state AI legislation that is found unconstitutional by SCOTUS.

What are your AI predictions for the next 12 months?